

Is General Artificial Intelligence Improvement Sustainable?

Introduction

From 2019 to 2024, global private investment in general artificial intelligence ballooned from around one billion dollars to almost 34 billion dollars. An underlying thesis of continued investment is (a) that those investments will produce returns (i.e., that general artificial intelligence companies will continue to grow in values and either dividends will pay out or the investor will be able to sell their shares in the company for a higher value than they paid) and (b) that those returns are significant enough to justify the initial cost of investment. Although global private investment has grown exponentially and the valuations of artificial intelligence companies like OpenAI or Anthropic have grown similarly exponentially over time, we argue that, on a basic technical level, this growth is unsustainable. General artificial intelligence requires compute to train: compute which requires microprocessors and money to purchase them. We draw a parallel to Moore's law, which is the thesis that the number of transistors per microprocessor follows an exponential curve. Moore's law couldn't have happened if transistors didn't decrease in price during its growth period. Transistors are to microprocessors what microprocessors are to compute. Even if investments continue at the pace they have—which is itself unlikely and unsustainable—more and more compute continues to be devoted to general artificial intelligence models with fewer and fewer returns per petaflop of compute. Compute demand vastly outpaces transistor price decline. Therefore, we argue that the current compute demands of general artificial intelligence are unsustainable, contributing to the hypothesis that general artificial intelligence investment itself is unsustainable.

Training Computation Over Time

Figure 1 shows training computation over time, starting in 1971. Figure 2 shows training computation over time, beginning in 2011.

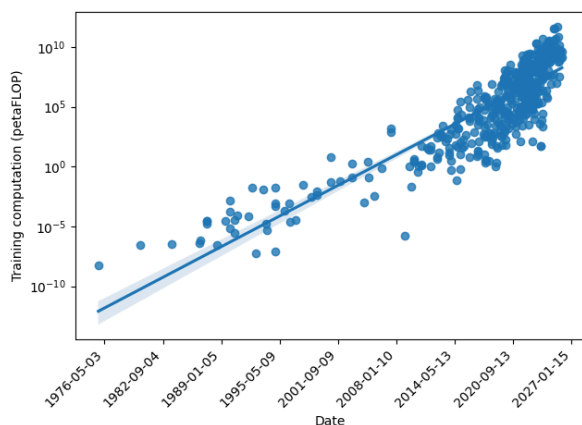


Figure 1: Training computation in petaFLOPS on a log scale against time starting in 1971

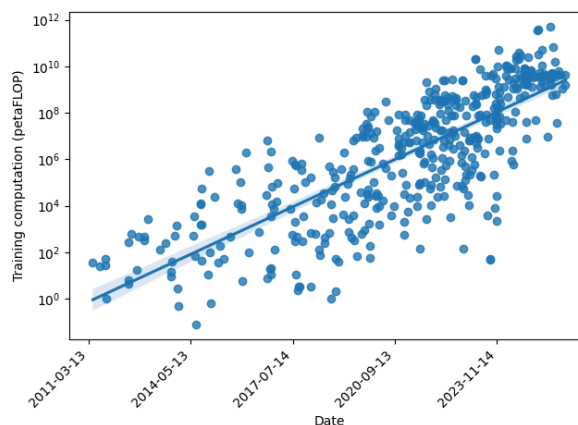


Figure 2: Training computation in petaFLOPS on a log scale against time starting in 2011

Both graphs demonstrate a clear exponential correlation between time and training computation.